Exercice 2 Aggregate function and Grouping

Q1: Count ( ) function
Write a SQL query to find the total number of employees in the company

A: SELECT COUNT (*) AS total employees
FROM sales

Total_employees
10

Q2: SUM Function
Write a SQL query to find the total salary paid to all employees in the IT department

A: SELECT sum (salary) AS total_salary
FROM employees
WHERE department = 'IT'

total_salary
22 000

Q3: AVG ( ) Function
Write a SQL query to calcutate the average salary of employees in the HR department

A: SELECT AVG (salary) AS average_HR_Salary
FROM employees
WHERE department = 'HR'

Average_HR_salary
49 500

Q4: MIN( ) AND MAX( ) function
write a SQL query to find the highest and the lowest salary in the company

A: SELECT
MIN (salary) AS lowest_salary,
MAX (salary) AS highest_salary
FROM employees

lowest_salary	highest_salary
48 000	62 000

Q5: Group by function
write a SQL query to group employees by department and display total salary in each department

A: SELECT department
SUM (salary) AS total_salary
FROM employees
GROUP BY department

Department	Total_salary
IT	220 000
HR	99 000
Finance	119 000
marketing	105 000

Q6 : Group BY and Count ( )
write a SQL query to count how many employees work in each city

A: SELECT City,
COUNT(*) AS total_employees
FROM employees
ORDER BY City

City	total_employee
New York	2
Chicargo	3
Los Angeles	2
San Fiancisco	2
Houston	1

Q7 : Group by and order by
write a SQL to group employees by department, calculate the avergge salary in each department and order the results in descending order of average salary.

A: SELECT department
AVG(salary) AS AVG_salary
FROM employees
GROUP BY department
ORDER BY avg_salary DESC;

Q8: HAVING STATEMENT
write a SQL query to find department where the total salary paid exceeds 100 000 (use group by and Having

A: SELECT department
SUM (salary) AS total_salary
FROM employee
GROUP BY department
HAVING SUM (salary) > 100 000

Department	total_salary
IT	220 000
Finance	119 000
Marketing	105 000

Q9: Combining Group by, Having and Order by
write SQL query to list cities where more than one employee works, order by the number of employees in descending order.
SELECT city
Count (*) AS employee_count
FROM employees
Group BY city
ORDER by employee_count Desc,

Q10: combining aggregate function
Write a SQL query to find the department with the highest average salary

A: SELECT department
AVG (salary) AS AVG_salary Desc
limit 1;

Department	AVG_salary
Finance	59 000